CLASS X : CHAPTER - 1
REAL NUMBERS

IMPORTANT FORMULAS & CONCEPTS

EUCLID'S DIVISION LEMMA

Given positive integers a and b, there exist unique integers q and r satisfying a = bq + r, where $0 \le r < b$.

Here we call 'a' as dividend, 'b' as divisor, 'q' as quotient and 'r' as remainder.

$\therefore$ Dividend = (Divisor x Quotient) + Remainder

If in Euclid's lemma r = 0 then b would be HCF of 'a' and 'b'.

NATURAL NUMBERS

Counting numbers are called natural numbers i.e. 1, 2, 3, 4, 5, ……. are natural numbers.

WHOLE NUMBERS

All counting numbers/natural numbers along with 0 are called whole numbers i.e. 0, 1, 2, 3, 4, 5 ……. are whole numbers.

INTEGERS

All natural numbers, negative of natural numbers and 0, together are called integers. i.e.

……. – 3, – 2, – 1, 0, 1, 2, 3, 4, …….. are integers.

ALGORITHM

An **algorithm** is a series of well defined steps which gives a procedure for solving a type of problem.

LEMMA

A **lemma** is a proven statement used for proving another statement.

EUCLID'S DIVISION ALGORITHM

Euclid's division algorithm is a technique to compute the Highest Common Factor (HCF) of two given positive integers. Recall that the HCF of two positive integers *a* and *b* is the largest positive integer *d* that divides both *a* and *b*.

To obtain the HCF of two positive integers, say c and d, with c > d, follow the steps below:

Step 1 : Apply Euclid's division lemma, to *c* and *d*. So, we find whole numbers, *q* and *r* such that $c = dq + r$, $0 \le r < d$.

Step 2 : If $r = 0$, *d* is the HCF of *c* and *d*. If $r \neq 0$ apply the division lemma to *d* and *r*.

Step 3 : Continue the process till the remainder is zero. The divisor at this stage will be the required HCF.

This algorithm works because HCF (*c*, *d*) = HCF (*d*, *r*) where the symbol HCF (*c*, *d*) denotes the HCF of *c* and *d*, etc.

The Fundamental Theorem of Arithmetic

Every composite number can be expressed (*factorised*) *as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.*

The prime factorisation of a natural number is unique, except for the order of its factors.

- ❖ HCF is the highest common factor also known as GCD i.e. greatest common divisor.
- ❖ LCM of two numbers is their least common multiple.
- ❖ Property of HCF and LCM of two positive integers 'a' and 'b':

- $HCF(a,b) \times LCM(a,b) = a \times b$
- $LCM(a,b) = \dfrac{a \times b}{HCF(a,b)}$
- $HCF(a,b) = \dfrac{a \times b}{LCM(a,b)}$

PRIME FACTORISATION METHOD TO FIND HCF AND LCM

HCF(a, b) = Product of the smallest power of each common prime factor in the numbers.
LCM(a, b) = Product of the greatest power of each prime factor, involved in the numbers.

RATIONAL NUMBERS

The number in the form of $\dfrac{p}{q}$ where 'p' and 'q' are integers and $q \neq 0$, e.g. $\dfrac{2}{3}, \dfrac{3}{5}, \dfrac{5}{7}, ...$

Every rational number can be expressed in decimal form and the decimal form will be either terminating or non-terminating repeating. e.g. $\dfrac{5}{2} = 2.5$ (Terminating), $\dfrac{2}{3} = 0.66666....$ *or* $0.\overline{6}$ (Non-terminating repeating).

IRRATIONAL NUMBERS

The numbers which are not rational are called irrational numbers. e.g. $\sqrt{2}, \sqrt{3}, \sqrt{5}, etc$.

- *Let p be a prime number. If p divides a2, then p divides a, where a is a positive integer.*
- If p is a positive integer which is not a perfect square, then $\sqrt{m}$ is an irrational, e.g. $\sqrt{2}, \sqrt{5}, \sqrt{6}, \sqrt{8}, ...etc$.
- If p is prime, then $\sqrt{p}$ is also an irrational.

RATIONAL NUMBERS AND THEIR DECIMAL EXPANSIONS

- *Let x be a rational number whose decimal expansion terminates. Then x can be expressed in the form $\dfrac{p}{q}$ where p and q are coprime, and the prime factorisation of q is of the form 2^n5^m, where n, m are non-negative integers.*

- *Let $x = \dfrac{p}{q}$ be a rational number, such that the prime factorisation of q is of the form 2^n5^m, where n, m are non-negative integers. Then x has a decimal expansion which terminates.*
- *Let $x = \dfrac{p}{q}$ be a rational number, such that the prime factorisation of q is not of the form 2^n5^m, where n, m are non-negative integers. Then, x has a decimal expansion which is non-terminating repeating* (*recurring*).

- *The decimal form of irrational numbers is non-terminating and non-repeating.*
- *Those decimals which are non-terminating and non-repeating will be irrational numbers. e.g. 0.20200200020002……. is a non-terminating and non-repeating decimal, so it irrational.*

MCQ WORKSHEET-I
CLASS X : CHAPTER - 1
REAL NUMBERS

1. A rational number between $\frac{3}{5}$ and $\frac{4}{5}$ is:

(a) $\frac{7}{5}$ (b) $\frac{7}{10}$ (c) $\frac{3}{10}$ (d) $\frac{4}{10}$

2. A rational number between $\frac{1}{2}$ and $\frac{3}{4}$ is:

(a) $\frac{2}{5}$ (b) $\frac{5}{8}$ (c) $\frac{4}{3}$ (d) $\frac{1}{4}$

3. Which one of the following is not a rational number:

(a) $\sqrt{2}$ (b) 0 (c) $\sqrt{4}$ (d) $\sqrt{-16}$

4. Which one of the following is an irrational number:

(a) $\sqrt{4}$ (b) $3\sqrt{8}$ (c) $\sqrt{100}$ (d) $-\sqrt{0.64}$

5. $3\frac{3}{8}$ in decimal form is:

(a) 3.35 (b) 3.375 (c) 33.75 (d) 337.5

6. $\frac{5}{6}$ in the decimal form is:

(a) $0.8\bar{3}$ (b) $0.8\overline{33}$ (c) $0.6\bar{3}$ (d) $0.6\overline{33}$

7. Decimal representation of rational number $\frac{8}{27}$ is:

(a) $0.\overline{296}$ (b) $0.29\bar{6}$ (c) $0.2\overline{96}$ (d) 0.296

8. 0.6666…….. in $\frac{p}{q}$ form is:

(a) $\frac{6}{99}$ (b) $\frac{2}{3}$ (c) $\frac{3}{5}$ (d) $\frac{1}{66}$

9. The value of $\left(\sqrt{5}+\sqrt{2}\right)\left(\sqrt{5}-\sqrt{2}\right)$ is:

(a) 10 (b) 7 (c) 3 (d) $\sqrt{3}$

10. $0.\overline{36}$ in $\frac{p}{q}$ form is:

(a) $\frac{6}{99}$ (b) $\frac{2}{3}$ (c) $\frac{3}{5}$ (d) none of these

MCQ WORKSHEET-II
CLASS X : CHAPTER - 1
REAL NUMBERS

1. $\sqrt{5}-3-2$ is

(a) a rational number (b) a natural number (c) equal to zero (d) an irrational number

2. Let x = $\frac{7}{20\times 25}$ be a rational number. Then x has decimal expansion, which terminates:

(a) after four places of decimal (b) after three places of decimal
(c) after two places of decimal (d) after five places of decimal

3. The decimal expansion of $\frac{63}{72\times 175}$ is

(a) terminating (b) non-terminating
(c) non termination and repeating (d) an irrational number

4. If HCF and LCM of two numbers are 4 and 9696, then the product of the two numbers is:

(a) 9696 (b) 24242 (c) 38784 (d) 4848

5. $\left(2+\sqrt{3}+\sqrt{5}\right)$ is :

(a) a rational number (b) a natural number (c) a integer number (d) an irrational number

6. If $\left(\frac{9}{7}\right)^3\times\left(\frac{49}{81}\right)^{2x-6}=\left(\frac{7}{9}\right)^9$, the value of x is:

(a) 12 (b) 9 (c) 8 (d) 6

7. The number .211 2111 21111….. is a

(a) terminating decimal (b) non-terminating decimal
(c) non termination and non-repeating decimal (d) none of these

8. If $(m)^n=32$ where m and n are positive integers, then the value of $(n)^{mn}$ is:

(a) 32 (b) 25 (c) 5^{10} (d) 5^{25}

9. The number $0.\overline{57}$ in the $\frac{p}{q}$ form $q\neq 0$ is

(a) $\frac{19}{35}$ (b) $\frac{57}{99}$ (c) $\frac{57}{95}$ (d) $\frac{19}{30}$

10. The number $0.5\overline{7}$ in the $\frac{p}{q}$ form $q\neq 0$ is

(a) $\frac{26}{45}$ (b) $\frac{13}{27}$ (c) $\frac{57}{99}$ (d) $\frac{13}{29}$

11. Any one of the numbers a, a + 2 and a + 4 is a multiple of:

(a) 2 (b) 3 (c) 5 (d) 7

12. If p is a prime number and p divides k^2, then p divides:

(a) $2k^2$ (b) k (c) 3k (d) none of these

MCQ WORKSHEET-III
CLASS X : CHAPTER - 1
REAL NUMBERS

1. π is
(a) a natural number (b) not a real number
(c) a rational number (d) an irrational number

2. The decimal expansion of π
(a) is terminating (b) is non terminating and recurring
(c) is non terminating and non recurring (d) does not exist.

3. Which of the following is not a rational number?
(a) $\sqrt{6}$ (b) $\sqrt{9}$ (c) $\sqrt{25}$ (d) $\sqrt{36}$

4. Which of the following is a rational number?
(a) $\sqrt{36}$ (b) $\sqrt{12}$ (c) $\sqrt{14}$ (d) $\sqrt{21}$

5. If a and b are positive integers, then HCF (a, b) x LCM (a, b) =
(a) a x b (b) a + b (c) a – b (d) a/b

6. If the HCF of two numbers is 1, then the two numbers are called
(a) composite (b) relatively prime or co-prime
(c) perfect (d) irrational numbers

7. The decimal expansion of $\frac{93}{1500}$ will be
(a) terminating (b) non-terminating (c) non-terminating repeating
(d) non-terminating non-repeating.

8. $\sqrt{3}$ is
(a) a natural number (b) not a real number
(c) a rational number (d) an irrational number

9. The HCF of 52 and 130 is
(a) 52 (b) 130 (c) 26 (d) 13

10. For some integer q, every odd integer is of the form
(a) q (b) q + 1 (c) 2q (d) none of these

11. For some integer q, every even integer is of the form
(a) q (b) q + 1 (c) 2q (d) none of these

12. Euclid's division lemma state that for any positive integers a and b, there exist unique integers q and r such that a = bq + r where r must satisfy
(a) $1 < r < b$ (b) $0 < r \leq b$ (c) $0 \leq r < b$ (d) $0 < r < b$

MCQ WORKSHEET-IV
CLASS X : CHAPTER - 1
REAL NUMBERS

1. A **……** is a proven statement used for proving another statement.
(a) axiom (b) theorem (c) lemma (d) algorithm

2. The product of non-zero rational ad an irrational number is
(a) always rational (b) always irrational (c) rational or irrational (d) one

3. The HCF of smallest composite number and the smallest prime number is
(a) 0 (b) 1 (c) 2 (d) 3

4. Given that HCF(1152, 1664) = 128 the LCM(1152, 1664) is
(a) 14976 (b) 1664 (c) 1152 (d) none of these

5. The HCF of two numbers is 23 and their LCM is 1449. If one of the numbers is 161, then the other number is
(a) 23 (b) 207 (c) 1449 (d) none of these

6. Which one of the following rational number is a non-terminating decimal expansion:
(a) $\frac{33}{50}$ (b) $\frac{66}{180}$ (c) $\frac{6}{15}$ (d) $\frac{41}{1000}$

7. A number when divided by 61 gives 27 quotient and 32 as remainder is
(a) 1679 (b) 1664 (c) 1449 (d) none of these

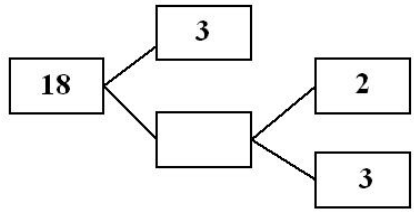
8. The product of L.C.M and H.C.F. of two numbers is equal to
(a) Sum of numbers (b) Difference of numbers
(c) Product of numbers (d) Quotients of numbers

9. L.C.M. of two co-prime numbers is always
(a) product of numbers (b) sum of numbers
(c) difference of numbers (d)none

10. What is the H.C.F. of two consecutive even numbers
(a) 1 (b)2 (c) 4 (d) 8

11. What is the H.C.F. of two consecutive odd numbers
(a) 1 (b) 2 (c) 4 (d) 8

12. The missing number in the following factor tree is
(a) 2 (b) 6 (c) 3 (d) 9



MCQ WORKSHEET-V
CLASS X : CHAPTER - 1
REAL NUMBERS

1. For some integer m, every even integer is of the form
(a) m (b) $m + 1$ (c) $2m$ (d) $2m + 1$

2. For some integer q, every odd integer is of the form
(a) q (b) $q + 1$ (c) $2q$ (d) $2q + 1$

3. $n^2 - 1$ is divisible by 8, if n is
(a) an integer (b) a natural number
(c) an odd integer (d) an even integer

4. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is
(a) 4 (b) 2 (c) 1 (d) 3

5. The largest number which divides 70 and 125, leaving remainders 5 and 8, respectively, is
(a) 13 (b) 65 (c) 875 (d) 1750

6. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$; x, y are prime numbers, then HCF (a, b) is
(a) xy (b) xy^2 (c) x^3y^3 (d) x^2y^2

7. If two positive integers p and q can be expressed as $p = ab^2$ and $q = a^3b$; a, b being prime numbers, then LCM (p, q) is
(a) ab (b) a^2b^2 (c) a^3b^2 (d) a^3b^3

8. The product of a non-zero rational and an irrational number is
(a) always irrational (b) always rational
(c) rational or irrational (d) one

9. The least number that is divisible by all the numbers from 1 to 10 (both inclusive) is
(a) 10 (b) 100 (c) 504 (d) 2520

10. The decimal expansion of the rational number $\frac{14587}{1250}$ will terminate after:
(a) one decimal place (b) two decimal places
(c) three decimal places (d) four decimal places

11. The decimal expansion of the rational number $\frac{33}{2^2.5}$ will terminate after
(a) one decimal place (b) two decimal places
(c) three decimal places (d) more than 3 decimal places

PRACTICE QUESTIONS
CLASS X : CHAPTER - 1
REAL NUMBERS

1. Write whether every positive integer can be of the form $4q + 2$, where q is an integer. Justify your answer.

2. "The product of two consecutive positive integers is divisible by 2". Is this statement true or false? Give reasons.

3. "The product of three consecutive positive integers is divisible by 6". Is this statement true or false"? Justify your answer.

4. Write whether the square of any positive integer can be of the form $3m + 2$, where m is a natural number. Justify your answer.

5. A positive integer is of the form $3q + 1$, q being a natural number. Can you write its square in any form other than $3m + 1$, i.e., $3m$ or $3m + 2$ for some integer m? Justify your answer.

6. Show that the square of an odd positive integer is of the form $8m + 1$, for some whole number m.

7. Show that the square of any positive integer is either of the form $4q$ or $4q + 1$ for some integer q.

8. Show that cube of any positive integer is of the form $4m$, $4m + 1$ or $4m + 3$, for some integer m.

9. Show that the square of any positive integer cannot be of the form $5q + 2$ or $5q + 3$ for any integer q.

10. Show that the square of any positive integer cannot be of the form $6m + 2$ or $6m + 5$ for any integer m.

11. Show that the square of any odd integer is of the form $4q + 1$, for some integer q.

12. If n is an odd integer, then show that $n^2 - 1$ is divisible by 8.

13. Prove that if x and y are both odd positive integers, then $x^2 + y^2$ is even but not divisible by 4.

14. Show that the square of an odd positive integer can be of the form $6q + 1$ or $6q + 3$ for some integer q.

15. Show that the cube of a positive integer of the form $6q + r$, q is an integer and $r = 0, 1, 2, 3, 4, 5$ is also of the form $6m + r$.

16. Prove that one and only one out of n, $n + 2$ and $n + 4$ is divisible by 3, where n is any positive integer.

17. Prove that one of any three consecutive positive integers must be divisible by 3.

18. For any positive integer n, prove that $n^3 - n$ is divisible by 6.

19. Show that one and only one out of n, $n + 4$, $n + 8$, $n + 12$ and $n + 16$ is divisible by 5, where n is any positive integer.

20. Show that the product of three consecutive natural numbers is divisble by 6.

21. Show that any positive odd integer is of the form 6q + 1 or 6q + 3 or 6q + 5 where $q \in Z$.

22. Show that any positive even integer is of the form 6q or 6q + 2 or 6q + 4 where $q \in Z$.

23. If a and b are two odd positive integers such that a > b, then prove that one of the two numbers $\frac{a+b}{2}$ and $\frac{a-b}{2}$ is odd and the other is even.

24. Use Euclid's division lemma to show that the cube of any positive integer is of the form 9m, 9m + 1 or 9m + 8.

25. Using Euclid's division algorithm to show that any positive odd integer is of the form 4q+1 or 4q+3, where q is some integer.

26. Use Euclid's division algorithm to find the HCF of 441, 567, 693.

27. Using Euclid's division algorithm, find the largest number that divides 1251, 9377 and 15628 leaving remainders 1, 2 and 3, respectively.

28. Using Euclid's division algorithm, find which of the following pairs of numbers are co-prime:
(i) 231, 396 (ii) 847, 2160

29. Show that 12^n cannot end with the digit 0 or 5 for any natural number *n*.

30. In a morning walk, three persons step off together and their steps measure 40 cm, 42 cm and 45 cm, respectively. What is the minimum distance each should walk so that each can cover the same distance in complete steps?

31. If LCM (480, 672) = 3360, find HCF (480,672).

32. Express 0.69 as a rational number in $\frac{p}{q}$ form.

33. Show that the number of the form 7^n, n∈N cannot have unit digit zero.

34. Using Euclid's Division Algorithm find the HCF of 9828 and 14742.

35. The numbers 525 and 3000 are both divisible only by 3, 5, 15, 25 and 75. What is HCF (525, 3000)? Justify your answer.

36. Explain why 3 × 5 × 7 + 7 is a composite number.

37. Can two numbers have 18 as their HCF and 380 as their LCM? Give reasons.

38. Without actual division find whether the rational number $\frac{1323}{\left(6^3 \times 35^2\right)}$ has a terminating or a non-terminating decimal.

39. Without actually performing the long division, find if $\frac{987}{10500}$ will have terminating or non-terminating (repeating) decimal expansion. Give reasons for your answer.

40. A rational number in its decimal expansion is 327.7081. What can you say about the prime factors of q, when this number is expressed in the form $\frac{p}{q}$? Give reasons.

41. Find the HCF of 81 and 237 and express it as a linear combination of 81 and 237.

42. Find the HCF of 65 and 117 and express it in the form 65m + 117n.

43. If the HCF of 210 and 55 is expressible in the form of 210x5 + 55y, find y.

44. If d is the HCF of 56 and 72, find x, y satisfying d = 56x + 72y. Also show that x and y are not unique.

45. Express the HCF of 468 and 222 as 468x + 222y where x, y are integers in two different ways.

46. Express the HCF of 210 and 55 as 210x + 55y where x, y are integers in two different ways.

47. If the HCF of 408 and 1032 is expressible in the form of 1032m – 408x5, find m.

48. If the HCF of 657 and 963 is expressible in the form of 657n + 963x(-15), find n.

49. A sweet seller has 420 kaju burfis and 130 badam burfis she wants to stack them in such a way that each stack has the same number, and they take up the least area of the tray. What is the number of burfis that can be placed in each stack for this purpose?

50. Find the largest number which divides 245 and 1029 leaving remainder 5 in each case.

51. Find the largest number which divides 2053 and 967 and leaves a remainder of 5 and 7 respectively.

52. Two tankers contain 850 litres and 680 litres of kerosene oil respectively. Find the maximum capacity of a container which can measure the kerosene oil of both the tankers when used an exact number of times.

53. In a morning walk, three persons step off together. Their steps measure 80 cm, 85 cm and 90 cm respectively. What is the minimum distance each should walk so that all can cover the same distance in complete steps?

54. Find the least number which when divided by 12, 16, 24 and 36 leaves a remainder 7 in each case.

55. The length, breadth and height of a room are 825 cm, 675 cm and 450 cm respectively. Find the longest tape which can measure the three dimensions of the room exactly.

56. Determine the smallest 3-digit number which is exactly divisible by 6, 8 and 12.

57. Determine the greatest 3-digit number exactly divisible by 8, 10 and 12.

58. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they change simultaneously at 7 a.m., at what time will they change simultaneously again?

59. Three tankers contain 403 litres, 434 litres and 465 litres of diesel respectively. Find the maximum capacity of a container that can measure the diesel of the three containers exact number of times.

60. Find the least number which when divided by 6, 15 and 18 leave remainder 5 in each case.

61. Find the smallest 4-digit number which is divisible by 18, 24 and 32.

62. Renu purchases two bags of fertiliser of weights 75 kg and 69 kg. Find the maximum value of weight which can measure the weight of the fertiliser exact number of times.

63. In a seminar, the number, the number of participants in Hindi, English and Mathematics are 60, 84 and 108, respectively. Find the minimum number of rooms required if in each room the same number of participants are to be seated and all of them being in the same subject.

64. 144 cartons of Coke cans and 90 cartons of Pepsi cans are to be stacked in a canteen. If each stack is of the same height and is to contain cartons of the same drink, what would be the greatest number of cartons each stack would have?

65. A merchant has 120 litres of oil of one kind, 180 litres of another kind and 240 litres of third kind. He wants to sell the oil by filling the three kinds of oil in tins of equal capacity. What would be the greatest capacity of such a tin?

66. Express each of the following positive integers as the product of its prime factors: (i) 3825 (ii) 5005 (iii) 7429

67. Express each of the following positive integers as the product of its prime factors: (i) 140 (ii) 156 (iii) 234

68. There is circular path around a sports field. Priya takes 18 minutes to drive one round of the field, while Ravish takes 12 minutes for the same. Suppose they both start at the same point and at the same time and go in the same direction. After how many minutes will they meet again at the starting point?

69. In a morning walk, three persons step off together and their steps measure 80 cm, 85 cm and 90 cm, respectively. What is the minimum distance each should walk so that each can cover the same distance in complete steps?

70. A circular field has a circumference of 360 km. Three cyclists start together and can cycle 48, 60 and 72 km a day, round the field. When will they meet again?

71. Find the smallest number which leaves remainders 8 and 12 when divided by 28 and 32 respectively.

72. Find the smallest number which when increased by 17 is exactly divisible by 520 and 468.

73. Find the greatest numbers that will divide 445, 572 and 699 leaving remainders 4, 5 and 6 respectively.

74. Find the greatest number which divides 2011 and 2423 leaving remainders 9 and 5 respectively

75. Find the greatest number which divides 615 and 963 leaving remainder 6 in each case.

76. Find the greatest number which divides 285 and 1249 leaving remainders 9 and 7 respectively.

77. Find the largest possible positive integer that will divide 398, 436, and 542 leaving remainder 7, 11, 15 respectively.

78. If d is the HCF of 30, 72, find the value of x & y satisfying d = 30x + 72y.

79. State Euclid's Division Lemma.

80. State the Fundamental theorem of Arithmetic.

81. Given that HCF (306, 657) = 9, find the LCM(306, 657).

82. Why the number 4^n, where n is a natural number, cannot end with 0?

83. Why is 5 x 7 x 11 + 7 is a composite number?

84. Explain why 7 x 11 + 13 + 13 and 7 x 6 x 5 x 4 x 3 x 2 x 1 + 5 are composite numbers.

85. In a school there are two sections – section A and section B of class X. There are 32 students in section A and 36 students in section B. Determine the minimum number of books required for their class library so that they can be distributed equally among students of section A or section B.

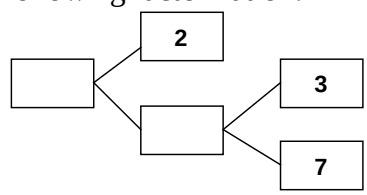
86. Determine the number nearest 110000 but greater than 100000 which is exactly divisible by each of 8, 15 and 21.

87. Three sets of English, Hindi and Mathematics books have to be stacked in such a way that all the books are stored topic wise and the height of each stack is the same. The number of English books is 96, the number of Hindi books is 240 and the number of Mathematics books is 336. Assuming that the books are of the same thickness, determine the number of stacks of English, Hindi and Mathematics books.

88. Using Euclid's division algorithm, find the HCF of 2160 and 3520.

89. Find the HCF and LCM of 144, 180 and 192 by using prime factorization method.

90. Find the missing numbers in the following factorization:



91. Find the HCF and LCM of 17, 23 and 37 by using prime factorization method.

92. If HCF(6, a) = 2 and LCM(6, a) = 60 then find the value of a.

93. If remainder of $\frac{(5m+1)(5m+3)(5m+4)}{5}$ is a natural number then find it.

94. A rational number $\frac{p}{q}$ has a non-terminating repeating decimal expansion. What can you say about q?

95. If $\frac{278}{2^m m}$ has a terminating decimal expansion and m is a positive integer such that $2 < m < 9$, then find the value of m.

96. Write the condition to be satisfied by q so that a rational number $\frac{p}{q}$ has a terminating expression.

97. If a and b are positive integers. Show that $\sqrt{2}$ always lies between $\frac{a}{b}$ and $\frac{a^2-2b^2}{b(a+b)}$.

98. Find two rational number and two irrational number between $\sqrt{2}$ and $\sqrt{3}$.

99. Prove that $5-2\sqrt{3}$ is an irrational number.

100. Prove that $15+17\sqrt{3}$ is an irrational number.

101. Prove that $\frac{2\sqrt{3}}{5}$ is an irrational number.

102. Prove that $7+3\sqrt{2}$ is an irrational number.

103. Prove that $2+3\sqrt{5}$ is an irrational number.

104. Prove that $\sqrt{2}+\sqrt{3}$ is an irrational number.

105. Prove that $\sqrt{3}+\sqrt{5}$ is an irrational number.

106. Prove that $7-2\sqrt{3}$ is an irrational number.

107. Prove that $3-\sqrt{5}$ is an irrational number.

108. Prove that $\sqrt{2}$ is an irrational number.

109. Prove that $7-\sqrt{5}$ is an irrational number

110. Show that there is no positive integer 'n' for which $\sqrt{n-1}+\sqrt{n+1}$ is rational.